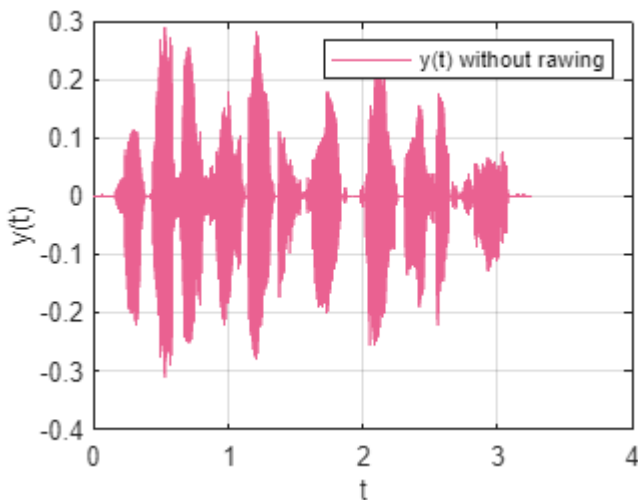


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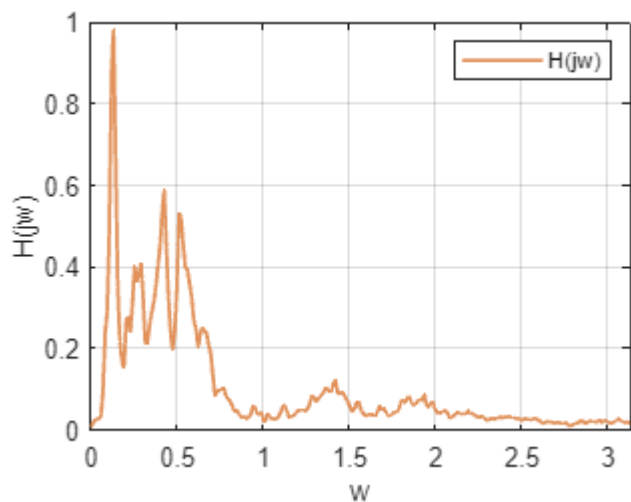
1. 题目要求根据已有语音信号生成语谱噪声，并画出源语音信号，频率响应和语谱噪声的频谱。

分析：先导入 BB 上自带的 C_01_01(1).wav 生源，运用已有声源生成语谱滤波器。而后自己构造一段白噪声，将白噪声经过滤波后即可得到语谱噪声。

```
clc
clear
[x,fs]=audioread('C_01_01(1).wav');
x = x';
t = linspace(0,length(x)/fs,length(x));
figure
plot(t,x,'Color',[234,97,145]/255);
xlabel('t')
ylabel('y(t)')
grid on
legend('y(t) without rawing')
```

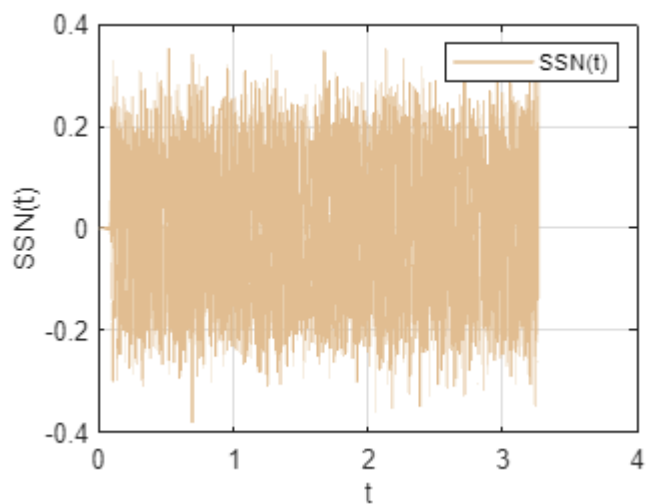


```
figure
y10 = repmat(x,1,10);
nfft=512;
noverlap=nfft/2;
window = hamming(nfft);
[P,w]=pwelch(y10>window,noverlap,nfft,fs);
b = filter(3000,w/(fs/2),sqrt(P/max(P)));
[H,W]=freqz(b,1);
plot(W,abs(H),"LineWidth",1.3,"Color",[225,145,89]/255)
xlabel('w')
ylabel('H(jw)')
grid on
legend('H(jw)')
```



以下代码用于生成白噪声。

```
figure
N = length(x);
noise = 1-2*rand(1,N);
SSN = filter(b,1,noise);
plot(t,SSN,"LineWidth",1.0,"Color",[225,189,145]/255)
xlabel('t')
ylabel('SSN(t)')
grid on
legend('SSN(t)')
```



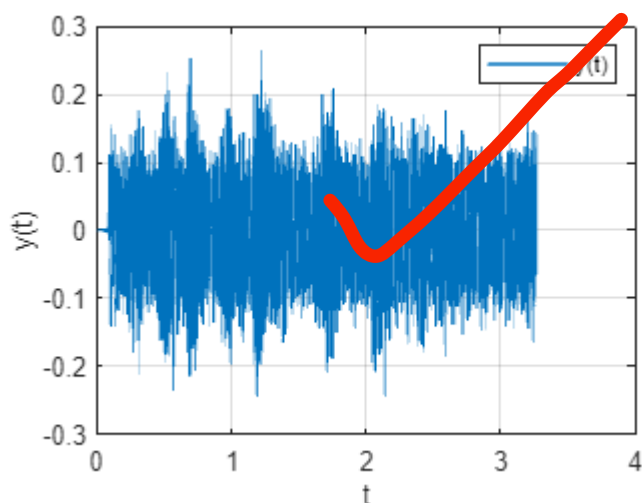
结论：原声音信号图像、滤波器频率响应、得到的语谱噪声如图所示。

2.题目要求在 SNR 为-5 分贝的情况下，在原语音信号中加入语谱噪声生成带噪声的信号，随后标准化带噪声信号 y 相对于原语言信号的 $x(t)$ 的能量。

分析： $SNR = -5 = 20 \cdot \lg(\text{norm}(x)/\text{norm}(SSN))$, $\text{norm}(SSN) = \text{norm}(x) \cdot 10^{(1/4)}$.

由此调整语谱噪声的能量，验证信噪比。将调整后的语谱噪声加入原信号中生成带噪音的信号，再调整带噪音的信号能量与原信号相同即可。

```
figure
SSNf = SSN/norm(SSN)*norm(x)*(10^0.25);
SNR = 20*log10(norm(x)/norm(SSNf));
y0 = x+SSNf;
y = y0/norm(y0)*norm(x);
plot(t,y,"LineWidth",1.0);
xlabel('t'),ylabel('y(t)'),grid on,legend("y(t)")
```

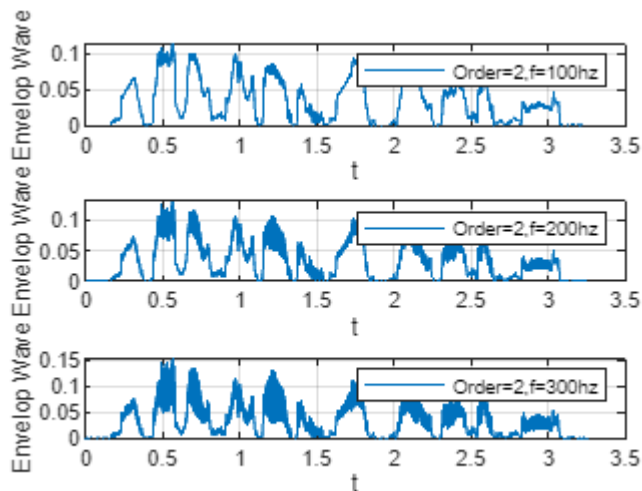


结论：得出的带噪声声音如图所示。

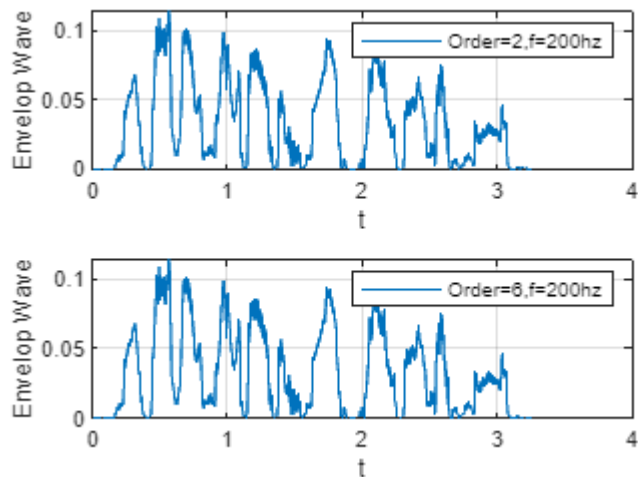
3.题目要求分别用二阶 100Hz,200Hz,300Hz 和六阶 200Hz 巴特沃斯低通滤波器获取包络，并比较差异.这里使用原信号的包络。

分析：按照题目所示阶数和截止频率分别构造巴特沃斯滤波器

```
figure
[b1,a1] = butter(2,100/(0.5*fs));
[b2,a2] = butter(2,200/(0.5*fs));
[b3,a3] = butter(2,300/(0.5*fs));
[b4,a4] = butter(6,200/(0.5*fs));
yb1 = filter(b1,a1,abs(x));
yb2 = filter(b2,a2,abs(x));
yb3 = filter(b3,a3,abs(x));
yb4 = filter(b4,a4,abs(x));
subplot(3,1,1)
plot(t,yb1),xlabel('t'),ylabel("Envelop Wave"),grid on, legend("Order=2,f=100hz");
subplot(3,1,2)
plot(t,yb2),xlabel('t'),ylabel("Envelop Wave"),grid on, legend("Order=2,f=200hz");
subplot(3,1,3)
plot(t,yb3),xlabel('t'),ylabel("Envelop Wave"),grid on, legend("Order=2,f=300hz");
```



```
figure
subplot(2,1,1)
plot(t,yb1,xlabel('t'),ylabel("Envelop Wave"),grid on, legend("Order=2,f=200hz");
subplot(2,1,2)
plot(t,yb1,xlabel('t'),ylabel("Envelop Wave"),grid on, legend("Order=6,f=200hz");
```



结论：由图知，

- ①阶数相同，截止频率不同，截止频率更高的滤波器可以保留高频信号，减少失真。
- ②当截止频率相同，阶数不同时，阶数更高的滤波器滤波效果更好。获取包络更加完整。

4. 录制一段声音信号并保存为.wav 文件

```
audiowrite('new_signal.wav',y0,fs);
```

漏过去的高频信号
少了

处理后保存的音频带有明显杂音，但原来内容仍可听清